CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Business Intelligence Report

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Project Summary

This project analyses customer shopping behaviour to uncover insights into sales performance, product effectiveness, and customer engagement patterns. The analysis identifies revenue concentration risks, subscription underperformance, gender imbalance in spending, and opportunities for product diversification and customer retention optimization.

Key findings show that while revenue is stable and diversified across items, the business is heavily dependent on a single product category and non-subscribed customers.

Project Objectives

 Analyze Customer Purchasing Behavior
To understand how customers shop by examining purchase frequency, spending patterns, and buying trends.

 Evaluate Product Performance
To identify the best-performing product categories and top-selling items that contribute most to overall revenue.

 Measure Business Performance
To track key metrics such as total revenue, average revenue per transaction, total customers, and average customer spending.

 Understand Customer Demographics
To analyze customer segments based on age group, gender, and location in order to identify the most valuable customer groups.

 Identify Sales Trends and Patterns
To explore seasonal sales trends and understand how different periods of the year influence purchasing behavior.

 Examine Payment Preferences
To determine the most commonly used payment methods among customers.

 Support Data-Driven Decision Making
To provide clear insights that help businesses improve marketing strategies, product planning, and inventory management.

 Improve Product and Inventory Strategy
To identify popular product attributes such as size and item type so businesses can optimize stock levels and meet customer demand.

DATASET OVERVIEW

Dataset contains information about customer demographics, purchasing patterns, product preferences, and engagement behavior. The dataset is designed to help analyze how different factors such as age, gender, product category, discounts, and subscription status influence customer spending and loyalty.

The dataset includes **3,900 customer records** with multiple attributes describing customer transactions and behaviors. Each record represents a purchase made by a customer and includes details about the product purchased, the amount spent, payment method, purchase frequency, and customer review ratings.

The data provides valuable insights into **customer segmentation, revenue drivers, shopping habits, and engagement levels**, making it suitable for building a **business intelligence dashboard** that supports data-driven decision making.

Key areas of analysis supported by this dataset include:

- Customer demographics and segmentation
- Sales and revenue performance
- Product category performance
- Customer engagement and loyalty
- Discount and promotion impact
- Payment method preferences
- Customer satisfaction through review ratings

Overall, this dataset enables analysts to explore patterns in consumer behavior, identify opportunities for improving customer retention, and support strategic decisions in marketing, sales, and product management.

Data Preparation

- Data loading
- Imported raw data into Power Bi Desktop
- Confirmed data structure and inspected column types

Data Cleaning & Transformation (Power Query)

- Checked for blank rows and columns (none identified)
- Created DAX measures and KPIs

Data Modelling

Created calculated columns and DAX measures for;

- Total Revenue
- Total Customers
- Revenue %
- Average Review Rating
- Average Purchase Value
- Average Previous Purchases
- No of Unique Items
- Repeat Customer %
- Age Column

Visualization

- Designed a clean, Business-Standard and user- friendly dashboard
- Included slicers for frequency of purchase, season, gender and location to enable dynamic analysis
- Added KPI cards for quick review
- Used column charts, bar charts, pie charts, donut chart and text boxes

Formatting & Branding

- A consistent theme for visual clarity was applied.
- Used layout alignment, colors and icons to enhance readability.

Sales Performance Analysis (Page 1)

Key KPIs

- **Total Revenue:** $233,081
- **Average Revenue per Transaction:** $59.76
- **Total Customers:** 3,900
- **Average Previous Purchases:** 25.35

Key Insights

Revenue by Category

Clothing generates approximately 45% of total revenue, creating category dependency risk.

Business Risk: Overreliance on a single category increases vulnerability to demand shifts.

Recommendation: Diversify revenue streams through product bundling and category expansion.

Revenue by Frequency of Purchase

Revenue distribution across purchase frequency types is relatively balanced, with quarterly and bi-weekly customers contributing slightly more.

Insight: Moderate-frequency buyers are major revenue stabilizers.

Recommendation: Develop targeted retention campaigns for these segments.

Top 5 Selling Items by Revenue

Top-selling items (Blouse, Shirt, Dress, Pants, Jewelry) generate similar revenue levels.

Insight: Healthy product distribution with no single-product dominance.

Product Performance Analysis (Page 2)

Key KPIs

- Total Revenue: $233,081
- Best Category: Clothing ($ 104.26k)
- Average Reviews Rating: 3.75
- Number of Unique Items: 25

Customer Satisfaction

Average rating of 3.75 indicates moderate satisfaction.

Risk: Long-term loyalty may decline without quality improvements.

Recommendation: Identify lowest-rated products and enhance quality control.

Seasonal Analysis – Revenue by Season and Category

Seasonal breakdown observations:

- Spring ($28k clothing) - Highest clothing sales
- Winter ($27k clothing) - Strong seasonal demand
- Fall ($26k clothing) – Stable
- Summer ($23k clothing) – Slight dip

Insights:

1. Clothing performs well year round.
2. Seasonal impact exists but is not extreme
3. Outerwear increases slightly in winter/ fall as expected

Recommendation:

1. Push seasonal promotions (e.g Outerwear in winter).
2. Offer summer fashion bundles to boost summer revenue.

Payment Behavior – Sales by Payment Method

Sales by Payment Method

1. Credit Card - $40k
2. Paypal - $40k
3. Cash - $39k
4. Debit Card - $39k
5. Venmo - $37k
6. Bank Transfer - $37k

Insights: Payment usage is evenly distributed. This means Customers are comfortable with multiple payment options and no strong dependency on a single payment channel.

Recommendation:

1. Offer small incentives for digital payments to reduce cash handling.
2. Partner with digital payment providers for cashback campaigns.

Size Distribution – Purchase by Size

- Size “M” significantly outperforms other sizes.

Recommendations:

- Maintain higher inventory levels for size M
- Avoid overstocking XL and S.

- Use sales/discounts to clear slower sizes.

Implication: Inventory planning should prioritize high-demand sizes.

Customer Behavior Analysis (Page 3)

Key KPIs

- **Total Customers:** 3,900
- **Average Spend per Customer:** $60
- **Total Orders:** 4,000
- **Repeat Customers:** 1%

Customer Loyalty and Retention - Repeat Customers

The Repeat Customer Percentage KPI provides insight into the level of customer loyalty and retention within the business. This metric measures the proportion of customers who make more than one purchase.

Insight: The Repeat Customers % shows a very low customer retention rate which indicates a problem in the business as higher repeat customer percentage indicates stronger customer satisfaction and brand loyalty, while a lower percentage suggests the business may rely heavily on acquiring new customers rather than retaining existing ones. And customer retention is critical for long-term profitability, as repeat customers typically contribute more consistent revenue and require lower acquisition costs compared to new customers.

Recommendations:

Promote Subscription or Membership Benefits by Highlighting benefits such as:

- Exclusive discounts
- Early access to new products
- Free shipping for subscribers

Expected Impact: Encourages long-term customer engagement and predictable revenue.

Use Customer Feedback and Reviews

Customer feedback helps identify reasons why customers may not return.

- Conduct customer satisfaction surveys
- Monitor product reviews
- Address common complaints quickly

Expected Impact: Improves products and services based on real customer needs.

Improve Customer Experience

Poor customer experience is a major reason why customers do not return. Improve:

- Delivery speed
- Customer support
- Website or app usability
- Product quality and packaging

Expected impact: Enhances customer satisfaction and encourages repeat purchases.

Age Distribution – Sale by Age Group

- Revenue is concentrated in the 25–54 age group.

  Insight: Core buyers are working professionals.

  Recommendation: Target marketing campaigns toward career-oriented demographics.

Gender Distribution – Sales by Gender

Sales by Gender

Male customers (67.74%) contribute over two-thirds of total revenue. The business appeared to attract more male buyers or higher male spending.

Recommendation:

1. Introduce marketing campaigns targeting female customers.
2. Expand female-oriented product offerings.

Subscription Effectiveness – Sales by Subscription Status

Sales by Subscription

More revenue comes from non-subscribed customers (73%).

The subscription program may be underutilized.

Recommendation:

1. Introduce exclusive benefits for subscribers.
2. Offer early product access, discount rewards, free-shipping.

By doing these could help increase loyalty and predictable profit.

Sales by Discount

More sales occur without discounts ($134k) and discounts still drive a significant portion of sales.

Insights: Customers are willing to purchase even without price reductions, suggesting:

1. Strong product value.
2. Brand or product loyalty.

Recommendation:

1. Use discounts strategically during slow sales period.
2. Avoid over-discounting to maintain profit margins.

Geographic Sales Distribution - Sales by Location

The Sales by Location analysis highlights how revenue is distributed across different regions. The visualization reveals that customer purchasing activity varies significantly by location, with some regions contributing a larger share of total sales than others.

It also shows that sales distribution across states is relatively balanced and no single state dominates overall revenue.

Montana - $5.8k, Illinois - $5.6K, California - $5.6k, Idaho - $5.6k, Nevada - $5.5k

The company has a diverse geographic customer base.

Recommendation:

- Identify high-potential states for regional marketing campaigns.
- Expand advertising in top-performing locations.

Strategic Recommendations

- Improve subscription value proposition and benefits.
- Target underrepresented customer segments.
- Enhance product quality control processes.
- Optimize inventory management based on size demand.
- Diversify product portfolio to reduce dependency risk.
- Focus on improving customer engagement through loyalty programs, personalized marketing, post-purchase incentives, and enhanced customer experience to improve customer retention and increase long-term customer lifetime value.

Tools Used

- Excel, Power Query and Power BI

Conclusion

- The business demonstrates stable revenue performance and diversified item-level sales; however, strategic improvements are required in subscription engagement, category diversification, and demographic expansion.
- By implementing targeted retention strategies and product optimization initiatives, the company can significantly improve customer lifetime value and revenue sustainability.